Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed.)
Solutions to the Exercises of Chapter 10: One-Dimensional Maps

Conventions. A map is $x_{n+1} = f(x_n)$. A fixed point x^* is stable if $|f'(x^*)| < 1$ and unstable if $|f'(x^*)| > 1$; a p-cycle has multiplier $(f^p)'(x_1) = \prod_i f'(x_i)$ and is stable if $|\text{multiplier}| < 1$. The logistic map is $x_{n+1} = rx_n(1 - x_n)$ and $x_m = \frac{1}{2}$ is its maximum point. I write "R" or "L" for an iterate lying to the right or left of x_m.

About the numerical exercises. Where an exercise asks for a computer experiment I ran it (double precision unless I say otherwise; high-precision root-finding with 30 digits for the superstable-cycle values) and the figures and numbers come from those runs. Where a conclusion rests only on numerics I say so. Facts I took from the literature rather than from my own runs are marked as such.

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1 Fixed Points and Cobwebs

10.1.1–10.1.8 *(Calculator experiments)*

Solution. Each map was iterated from several starting values. Summary ("unstable"/"stable" refer to the fixed point named):

map	fixed points	behavior
$\sqrt{x}$ (10.1.1)	0 (unstable, $f' = \infty$), 1 (stable, $f' = \frac{1}{2}$)	Every $x_0 > 0$ converges monotonically to 1 (e.g. $0.3 \to 0.548 \to 0.740 \to \dots$, $100 \to 10 \to 3.16 \to \dots$).
x^3 (10.1.2)	0 (superstable), ± 1 ($f' = 3$, unstable)	$\lvert x_0\rvert < 1 \to 0$ extremely fast ($0.9 \to 0.729 \to 0.387 \to 0.058 \to 2 \times 10^{-4} \to \cdots$); $\lvert x_0\rvert > 1 \to \pm\infty$ (calculator overflows after ~ 6 steps).
e^x (10.1.3)	none: $e^x > x$ for all x	Any x_0 gives $x_1 > 0$, then the sequence increases without bound (overflow within ~ 5 steps).
$\ln x$ (10.1.4)	none on $\mathbb{R}$; complex $x^* = 0.318 \pm 1.337i$ with $\lvert f'\rvert = 1/\lvert x^*\rvert = 0.73 < 1$	Real orbits decrease until some iterate is ≤ 0 and then ln is undefined (e.g. $10 \to 2.30 \to 0.834 \to -0.181 \to$ error). Only in the complex plane do orbits converge, to the complex fixed point.
$\cot x$ (10.1.5)	$x^* = 0.8603$, $f' = -(1+x^{*2}) = -1.74$ (unstable)	$\lvert f'\rvert = \csc^2 x \ge 1$ everywhere, so no stable cycle can exist. Orbits wander erratically (e.g. $1 \to 0.642 \to 1.337 \to 0.238 \to 4.124 \to \cdots$).
$\tan x$ (10.1.6)	0 ($f' = 1$, repelling since $\tan x \approx x + x^3/3$); $x^* = 4.4934, \dots$ ($f' = 21.2$)	$\lvert f'\rvert = \sec^2 x \ge 1$, so again no attractor; orbits are erratic and unbounded excursions occur ($1 \to 1.557 \to 74.7 \to -0.864 \to \cdots$).
$\sinh x$ (10.1.7)	0 only ($f' = 1$)	$\sinh x > x$ for $x > 0$: orbits increase monotonically, and since there is no larger fixed point they diverge to $\pm\infty$ (very slowly for tiny x_0, because $\sinh x \approx x + x^3/6$). The origin repels.
$\tanh x$ (10.1.8)	0 ($f' = 1$)	$0 < \tanh x < x$ for $x > 0$: orbits decrease monotonically and converge to 0, but only algebraically: $x_{n+1} \approx x_n - x_n^3/3$ gives $x_n \approx \sqrt{3/(2n)}$ (numerically $x_n\sqrt{2n/3} = 0.9999$ at $n = 10^4$).

Note the pattern for the marginal ($f' = 1$) fixed points: the sign of the cubic term decides stability (tan, sinh repel, tanh attracts). ■

10.1.9 *($x_{n+1} = 2x_n/(1 + x_n)$)*

Solution. Fixed points: $x(1 + x) = 2x \Rightarrow x = 0, 1$. $f'(x) = 2/(1 + x)^2$: $f'(0) = 2$ (unstable), $f'(1) = \frac{1}{2}$ (stable). A useful identity is

$$x_{n+1} - 1 = \frac{x_n - 1}{1 + x_n}.$$

Positive x_0: $\lvert x_{n+1} - 1\rvert = \lvert x_n - 1\rvert/(1 + x_n) < \lvert x_n - 1\rvert$, so $x_n \to 1$ monotonically.
Negative x_0: for $-1 < x < 0$ we have $x_{n+1} = x \cdot \frac{2}{1+x} < x$ (the factor exceeds 2), so the orbit decreases until it passes below -1 (it cannot converge since there is no fixed point in $(-1, 0)$). For $x < -1$, $x_{n+1} = 2 + \frac{2}{\lvert 1+x\rvert} > 2$, so the next iterate is positive and the orbit then converges to 1. The only exceptions are the countably many starting values that land exactly on $x = -1$ (e.g. $x_0 = -\frac{1}{3} \mapsto -1$), where the map is undefined. So $x^* = 1$ attracts every initial condition except $x_0 = 0$ (fixed) and those exceptional points. ■

10.1.10 *($x_{n+1} = 1 + \frac{1}{2}\sin x_n$)*

Solution. Let $g(x) = 1 + \frac{1}{2}\sin x - x$. Then $g'(x) = \frac{1}{2}\cos x - 1 \le -\frac{1}{2} < 0$, so g is strictly decreasing; $g(0) = 1 > 0$ and $g(2) < 0$, so g has exactly one zero: a unique fixed point $x^* \approx 1.4987$. Stability:

$|f'(x^*)| = \frac{1}{2}|\cos x^*| \le \frac{1}{2} < 1$ (numerically 0.036). So it is stable (indeed globally attracting, since $|f'| \le \frac{1}{2}$ everywhere makes f a contraction). ■

10.1.11 *(Cubic map $x_{n+1} = 3x_n - x_n^3$)*

Solution. **(a)** $3x - x^3 = x \Rightarrow x(2 - x^2) = 0$: $x^* = 0, \pm\sqrt{2}$. $f'(x) = 3 - 3x^2$: $f'(0) = 3$, $f'(\pm\sqrt{2}) = -3$. All three are unstable.
(b) $x_0 = 1.9$: 1.9, −1.159, −1.920, 1.319, 1.662, 0.394, 1.120, 1.955, −1.607, −0.673, … The orbit stays in $[-2, 2]$ and jumps around irregularly (bounded, aperiodic-looking).
(c) $x_0 = 2.1$: $2.1 \to -2.961 \to 17.08 \to -4.8 \times 10^3 \to \cdots$ and $|x_n| \to \infty$ (superexponentially).
(d) f has critical points $x = \pm 1$ with $f(\pm 1) = \pm 2$, and $f(\pm 2) = \mp 2$, so f maps $[-2, 2]$ into itself: $|x_0| \le 2 \Rightarrow |x_n| \le 2$ for all n (bounded). If $|x| > 2$ then $|f(x)| = |x|(x^2 - 3) > |x|$ since $x^2 - 3 > 1$; the ratio $|x_{n+1}|/|x_n| = x_n^2 - 3 > 1$ itself increases, so $|x_n| \to \infty$. The interval $[-2, 2]$ is the trapping region. ■

10.1.12 *(Newton's method)*

Solution. **(a)** $g = x^2 - 4$, $g' = 2x$: $f(x) = x - \dfrac{x^2 - 4}{2x} = \dfrac{x}{2} + \dfrac{2}{x}$.
(b) $x = \frac{x}{2} + \frac{2}{x} \Leftrightarrow \frac{x}{2} = \frac{2}{x} \Leftrightarrow x^2 = 4$, so $x^* = \pm 2$.
(c) $f'(x) = \frac{1}{2} - 2/x^2$, so $f'(\pm 2) = 0$: superstable.
(d) From $x_0 = 1$: 1, 2.5, 2.05, 2.0006098, 2.0000000929, 2.00000000000000 (errors 1, 0.5, 0.05, 6.1×10^{-4}, 9.3×10^{-8}, $\sim 10^{-15}$): the number of correct digits roughly doubles each step, since $e_{n+1} \approx e_n^2/4$. ■

10.1.13 *(Newton's method and superstability)*

Solution. Let $f(x) = x - g(x)/g'(x)$ with $g \in C^2$. Then

$$f'(x) = 1 - \frac{g'^2 - gg''}{g'^2} = \frac{g(x)g''(x)}{g'(x)^2}.$$

If $g(x^*) = 0$ and $g'(x^*) \neq 0$ (a simple root; g'' finite near x^*), then $f'(x^*) = 0$: *every simple root of g is a superstable fixed point of the Newton map.* Conversely a fixed point of f needs $g(x)/g'(x) = 0$, i.e. $g(x) = 0$ (where $g' \neq 0$). Since $f'(x^*) = 0$, Taylor's theorem gives $e_{n+1} \approx \frac{1}{2}f''(x^*)e_n^2 = \frac{g''(x^*)}{2g'(x^*)}e_n^2$: quadratic convergence. (If the root is multiple, $g'(x^*) = 0$, then $f'(x^*) = 1 - 1/m \neq 0$ for a root of multiplicity m, and convergence is only geometric.) ■

10.1.14 *($x_{n+1} = -\sin x_n$)*

Solution. Because $|\sin x| \le |x|$ with equality only at $x = 0$, we have $|x_{n+1}| = |\sin x_n| \le |x_n|$: $|x_n|$ is non-increasing and bounded below, so it converges to some $L \ge 0$. Along a subsequence $x_{n_k} \to \pm L$, and continuity gives $L = |\sin L|$; since $|\sin x| < |x|$ for $x \neq 0$ this forces $L = 0$. Hence $x_n \to 0$ for *every* x_0: $x^* = 0$ is globally stable. (Cobweb picture: the graph of $-\sin x$ lies between the lines $y = x$ and $y = -x$, so the cobweb spirals into the origin; the fixed point is non-hyperbolic since $f'(0) = -1$, and convergence is algebraic, $x_n \sim n^{-1/2}$.) ■

2 Logistic Map: Numerics

10.2.1 *(Logistic map for all real x, $r > 1$)*

Solution. **(a)** If $x_n > 1$ then $1 - x_n < 0$, so $x_{n+1} = rx_n(1 - x_n) < 0$. For $x < 0$ we have $x_{n+1}/x_n = r(1 - x_n) = r(1 + |x_n|) \geq r > 1$, hence $|x_{n+1}| \geq r|x_n|$ and $x_{n+1} < x_n$: the iterates are negative, decreasing, and growing geometrically at least, so $x_n \to -\infty$ (population extinction in the application).
(b) Orbits that leave $[0, 1]$ diverge, so only $x \in [0, 1]$ is of interest. f maps $[0, 1]$ into itself iff its maximum $r/4 \leq 1$, i.e. $r \leq 4$ (for $r > 4$ some points of $[0, 1]$ are mapped above 1 and escape). Also $r < 0$ is equivalent to $r > 0$ after a change of variables, so $r \in [0, 4]$ is the interesting range. ■

10.2.2 *($x^* = 0$ globally stable for $0 \leq r \leq 1$)*

Solution. For $x \in (0, 1]$ and $r \leq 1$: $f(x) = rx(1 - x) < x$, so the graph lies strictly below the diagonal. The cobweb is a staircase descending to the origin: x_n is decreasing and bounded below by 0, hence convergent, and the limit is a fixed point, which must be 0 (the only fixed point in $[0, 1]$ for $r \leq 1$). Also $f([0, 1]) \subset [0, r/4] \subset [0, 1]$, so orbits starting in $[0, 1]$ stay there. (At $r = 1$ the convergence is only algebraic, $x_n \sim 1/n$.) Combined with 10.2.1, 0 attracts all x_0 in $[0, 1]$. ■

10.2.3 *(Orbit diagram of the logistic map)*

Solution. Features visible: stable fixed point $x^* = 1 - 1/r$ for $1 < r < 3$; flip at $r = 3$; the 2-cycle

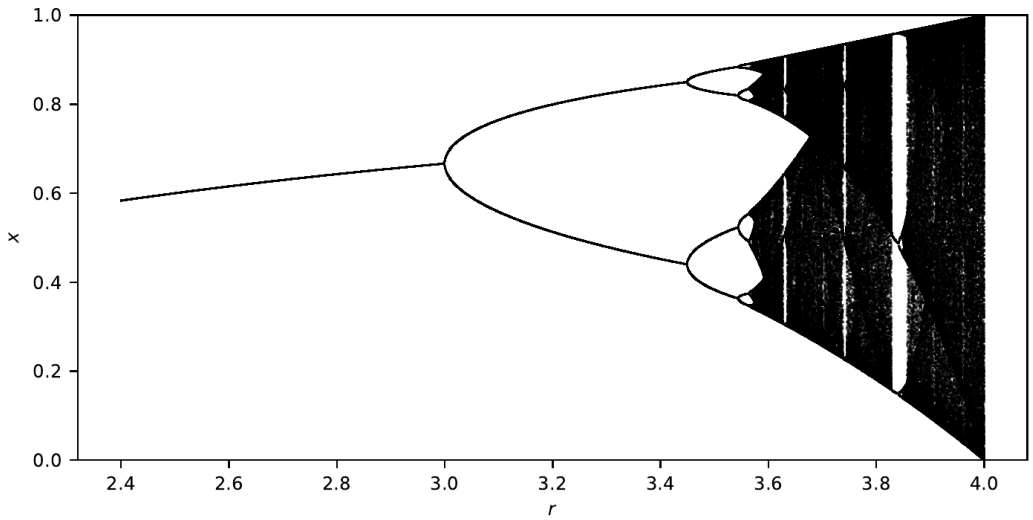


Figure 1: Logistic-map orbit diagram, $2.4 \leq r \leq 4$ (1500 transient iterations, 300 plotted).

for $3 < r < 1 + \sqrt{6} \approx 3.449$; the cascade of doublings accumulating at $r_\infty \approx 3.5700$; chaos with periodic windows beyond (the period-3 window starts at $r = 1 + \sqrt{8} \approx 3.8284$, see 10.4.10); and the full interval $[0, 1]$ filled at $r = 4$. ■

10.2.4–10.2.8 *(Orbit diagrams for other maps)*

Solution. **10.2.4** $x_{n+1} = x_n e^{r(1-x_n)}$ (Ricker): the fixed point $x^* = 1$ has $f'(1) = 1 - r$, stable for $0 < r < 2$ and losing stability at $r = 2$ by period doubling; then a doubling cascade and chaos, with periodic windows (the standard route).
10.2.5 $x_{n+1} = e^{-rx_n}$: unique fixed point $x^* = W(r)/r$ with $f'(x^*) = -rx^* = -W(r)$, stable for

$r < e$. At $r = e$ it flips to a 2-cycle that remains stable for *all* $r > e$ (approaching $\{0, 1\}$ as $r \to \infty$): "one period-doubling and the show is over". This can be explained: f is monotone decreasing, so f^2 is increasing, and orbits of an increasing map are monotone, so only fixed points and 2-cycles are possible.

10.2.6 $x_{n+1} = r \cos x_n$: fixed point at small r, then period doubling with chaos and periodic windows; the attractor grows like $|x| \le r$.

10.2.7 $x_{n+1} = r \tan x_n$: $x^* = 0$ is stable for $r < 1$ ($f'(0) = r$; at $r = 1$ it is a subcritical pitchfork since $\tan x \approx x + x^3/3$: two unstable fixed points exist for $r < 1$ and merge into the origin). For $r > 1$ the plot fills the whole window (the map has infinitely many singularities, so the attractor is unbounded and its structure is a "nasty mess").

10.2.8 $x_{n+1} = rx_n - x_n^3$: pitchfork at $r = 1$ to $x^* = \pm\sqrt{r-1}$, then (by the odd symmetry) each branch period-doubles at $r = 2$ and goes chaotic *independently*, giving a symmetric pair of attractors (the two plotted colors/initial signs); at larger r (about 2.6 on the picture, a rough reading) they merge into one symmetric attractor. See also 10.3.6. ■

3 Logistic Map: Analysis

10.3.1 *(Superstable fixed point)*

Solution. $f'(x) = r(1-2x) = 0$ at $x = \frac{1}{2}$. We need $\frac{1}{2}$ to be a fixed point: $x^* = 1 - \frac{1}{r} = \frac{1}{2} \Rightarrow r = 2$. So the fixed point is superstable at $\boxed{r = 2}$. ■

10.3.2 *(Superstable 2-cycle)*

Solution. **(a)** The multiplier is $f'(p)f'(q) = r^2(1-2p)(1-2q)$, which vanishes iff $p = \frac{1}{2}$ or $q = \frac{1}{2}$. **(b)** Take $p = \frac{1}{2}$: $q = f(p) = r/4$ and we need $f(q) = p$: $r \cdot \frac{r}{4}\left(1 - \frac{r}{4}\right) = \frac{1}{2} \Rightarrow r^2(4-r) = 8 \Rightarrow r^3 - 4r^2 + 8 = 0 \Rightarrow (r-2)(r^2 - 2r - 4) = 0$. The root $r = 2$ is the superstable fixed point of 10.3.1 (there $q = p = \frac{1}{2}$, not a 2-cycle); the other roots are $1 \pm \sqrt{5}$. Hence the superstable 2-cycle occurs at $\boxed{r = 1 + \sqrt{5} \approx 3.2361}$. ■

10.3.3 *($x_{n+1} = rx_n/(1 + x_n^2)$, $r > 0$)*

Solution. *Fixed points:* $x^* = 0$ for all r, and $1 + x^2 = r \Rightarrow x^* = \pm\sqrt{r-1}$ for $r > 1$. With $f'(x) = r(1 - x^2)/(1 + x^2)^2$: $f'(0) = r$ (stable for $r < 1$, unstable for $r > 1$), and at $x^{*2} = r - 1$, $f' = \dfrac{r(2-r)}{r^2} = \dfrac{2-r}{r}$, with $|(2-r)/r| < 1 \iff r > 1$. So the nonzero fixed points are stable for *every* $r > 1$: a supercritical pitchfork at $r = 1$ and nothing else.

No 2-cycles. Suppose $y = f(x)$, $x = f(y)$ with $x \ne y$: $y(1 + x^2) = rx$ and $x(1 + y^2) = ry$. Subtracting gives $(x - y)(xy - 1 - r) = 0$; adding gives $(x + y)(1 + xy - r) = 0$. For $x \ne y$ the first gives $xy = 1 + r$, which is incompatible with $xy = r - 1$, so the second forces $y = -x$; then $xy = -x^2 = 1 + r > 0$ is impossible. Hence no 2-cycle exists for any $r > 0$.

Consequences. f is continuous and maps the interval $[-r/2, r/2]$ into itself. By Sharkovskii's theorem a continuous interval map with a cycle of any period $n \ge 2$ also has a cycle of period 2, so there are **no periodic solutions other than fixed points**, and in particular no period-3 and hence no chaos. (An interval map with no 2-cycle has all orbits converging to fixed points.) Thus: $r \le 1$: all orbits $\to 0$; $r > 1$: $x_0 > 0 \to +\sqrt{r-1}$, $x_0 < 0 \to -\sqrt{r-1}$. ■

10.3.4 *(Quadratic map $x_{n+1} = x_n^2 + c$)*

Solution. **(a)** $x^2 - x + c = 0 \Rightarrow x_\pm^* = \frac{1\pm\sqrt{1-4c}}{2}$, existing for $c \le \frac{1}{4}$. $f'(x_\pm^*) = 2x_\pm^* = 1 \pm \sqrt{1-4c}$. Thus x_+^* is unstable ($f' \ge 1$) and x_-^* is stable iff $|1-\sqrt{1-4c}| < 1 \iff \sqrt{1-4c} < 2 \iff c > -\frac{3}{4}$. So x_-^* is stable for $-\frac{3}{4} < c < \frac{1}{4}$.
(b) At $c = \frac{1}{4}$ the two fixed points collide ($f' = 1$) and annihilate: a **saddle-node (tangent) bifurcation**. At $c = -\frac{3}{4}$, $f'(x_-^*) = -1$: a **flip (period-doubling) bifurcation**, supercritical (see (c)).
(c) $f^2(x) - x = (x^2+c)^2 + c - x = (x^2 - x + c)(x^2 + x + c + 1)$, so the 2-cycle satisfies $x^2 + x + c + 1 = 0$: $p, q = \frac{-1\pm\sqrt{-3-4c}}{2}$, which exists for $c < -\frac{3}{4}$ (born at the flip). Since $p + q = -1$, $pq = c + 1$, the multiplier is $f'(p)f'(q) = 4pq = 4(c+1)$. The 2-cycle is stable iff $|4(c+1)| < 1 \iff -\frac{5}{4} < c < -\frac{3}{4}$, and superstable ($pq = 0$) at $\boxed{c = -1}$, where the cycle is $\{0, -1\}$.
(d) Bifurcation diagram (solid = stable, dashed = unstable; grey = numerically computed orbits):
■

10.3.5 *(Conjugacy of logistic and quadratic maps)*

Solution. Put $x = ay + b$ in $x_{n+1} = rx_n(1 - x_n)$:

$$y_{n+1} = \frac{r}{a}(ay_n + b)(1 - b - ay_n) - \frac{b}{a} = -ra\, y_n^2 + r(1 - 2b)\, y_n + \frac{rb(1-b) - b}{a}.$$

Matching $y_{n+1} = y_n^2 + c$: the linear term vanishes if $b = \frac{1}{2}$; then $-ra = 1$ gives $a = -1/r$, and

$$\boxed{x = \tfrac{1}{2} - \frac{y}{r}, \qquad c = \frac{r}{2} - \frac{r^2}{4} = \frac{r(2-r)}{4}.}$$

Checks: $r = 4 \leftrightarrow c = -2$ (the end of chaos for the quadratic map), $r = 1 \leftrightarrow c = \frac{1}{4}$, and the logistic flip $r = 3 \leftrightarrow c = -\frac{3}{4}$ (the flip of 10.3.4). ■

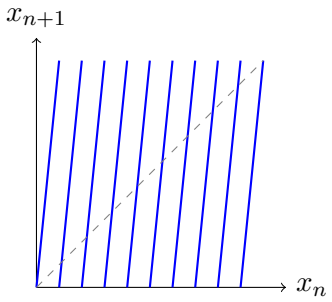
10.3.6 *(Cubic map $x_{n+1} = rx_n - x_n^3$)*

Solution. **(a)** Fixed points: $x^* = 0$ (all r) and $x^* = \pm\sqrt{r-1}$ (for $r > 1$). With $f' = r - 3x^2$: $f'(0) = r$, so 0 is stable for $-1 < r < 1$; $f'(\pm\sqrt{r-1}) = 3 - 2r$, so these are stable for $1 < r < 2$ (flip at $r = 2$).
(b) From $f(p) = q$, $f(q) = p$: $rp - p^3 = q$, $rq - q^3 = p$. Adding: $(r-1)(p+q) = p^3 + q^3 \Rightarrow (p+q)(p^2 - pq + q^2 - r + 1) = 0$. Subtracting: $(r+1)(p-q) = p^3 - q^3 \Rightarrow (p-q)(p^2 + pq + q^2 - r - 1) = 0$. For a genuine 2-cycle $p \ne q$, so $p^2 + pq + q^2 = r + 1$, and either
(i) $q = -p$: then $p^2 = r + 1$ (the cycle $\{\pm\sqrt{r+1}\}$, i.e. $x^2 - r - 1 = 0$), or
(ii) $p^2 - pq + q^2 = r - 1$: subtracting from $p^2 + pq + q^2 = r + 1$ gives $pq = 1$, and then $p^2 + q^2 = r$, so $p^2 + p^{-2} = r$, i.e. $x^4 - rx^2 + 1 = 0$.
Together with the fixed points $x(x^2 - r + 1) = 0$ these account for all nine roots of $f^2(x) = x$: $x(x^2 - r + 1)(x^2 - r - 1)(x^4 - rx^2 + 1) = 0$. The cycles (ii) require $r \ge 2$ (real roots: $r^2 - 4 \ge 0$) and, for $r > 2$, are the pairs $\{p, 1/p\}$ and $\{-p, -1/p\}$ with $p^2 = \frac{r+\sqrt{r^2-4}}{2}$.
(c) (i) $\{\pm a\}$, $a^2 = r + 1$: $f'(\pm a) = r - 3(r+1) = -2r - 3$, multiplier $(2r+3)^2$. It exists for $r > -1$ (born from the origin at $r = -1$ where $f'(0) = -1$) and its multiplier exceeds 1 for all $r > -1$: it is *always unstable*, so the flip at $r = -1$ is **subcritical**.
(ii) $\{p, 1/p\}$: multiplier $(r - 3p^2)(r - 3p^{-2}) = r^2 - 3r(p^2 + p^{-2}) + 9 = 9 - 2r^2$. It is stable iff $|9 - 2r^2| < 1 \iff 2 < r < \sqrt{5} \approx 2.236$. At $r = 2$ it emerges from the fixed point $x^* = 1$ (a

supercritical flip, consistent with $3 - 2r = -1$), and at $r = \sqrt{5}$ its multiplier is -1 (another flip, creating a 4-cycle).
(d) ■

10.3.7 *(Decimal shift map)*

Solution. **(a)** The graph is ten parallel line segments of slope 10, each rising from 0 to 1 over an interval of length 0.1 (a sawtooth):



(b) Fixed points: $10x - k = x \Rightarrow x = k/9$, $k = 0, 1, \ldots, 8$ (and $x = 1 \equiv 0$). In decimals these are 0 and the constant strings $0.\overline{1}$, $0.\overline{2}, \ldots, 0.\overline{8}$.
(c) Writing $x = 0.d_1d_2d_3\ldots$, the map drops the first digit: $x_{n+1} = 0.d_2d_3\ldots$. So x has period p iff its digits repeat with period p, i.e. $x = k/(10^p - 1)$. An explicit period-p point is $x = \dfrac{1}{10^p - 1} = 0.\overline{0\cdots01}$ ($p-1$ zeros then a 1; for $p > 1$ it is not a fixed point because the shifts $0.0\cdots01\,0\cdots$ are distinct). All periodic points are unstable: each step multiplies deviations by $|f'| = 10$, so the multiplier of a p-cycle is $10^p > 1$.
(d) Rational numbers have eventually periodic decimal expansions, hence eventually periodic orbits. An irrational x_0 (e.g. $\pi - 3$, $\sqrt{2} - 1$) has a non-repeating expansion, so its orbit is never periodic: there are infinitely many (indeed uncountably many) aperiodic orbits.
(e) Two points that agree in their first m digits and then differ, $|x - x'| \approx 10^{-m}$, satisfy $|x_n - x'_n| = 10^n|x - x'| \bmod 1$: the separation is multiplied by 10 each step until it is of order one after about m steps. Nearby orbits separate exponentially: the Liapunov exponent is $\ln 10 > 0$ (see 10.5.2). That is sensitive dependence. ■

10.3.8 *(Dense orbit for the decimal shift)*

Solution. Build x_0 by writing a decimal point and then concatenating *all* finite digit strings, in order of increasing length:

$$x_0 = 0.\,0\,1\,2\,3\,4\,5\,6\,7\,8\,9\;00\,01\,02\cdots99\;000\,001\cdots999\;0000\cdots$$

Given any $p \in [0, 1]$ and $\varepsilon > 0$, choose k with $10^{-k} < \varepsilon$ and let $p_1\cdots p_k$ be the first k digits of p. This block appears somewhere in x_0's expansion, beginning at digit $N + 1$ say. After N shifts, $x_N = 0.p_1\cdots p_k\ldots$, so $|x_N - p| < 10^{-k} < \varepsilon$. Hence the orbit is dense. ■

10.3.9 *(Binary shift map)*

Solution. Write $x = 0.b_1b_2b_3\ldots$ in binary; $x \mapsto 2x \bmod 1$ drops the first bit: $x_{n+1} = 0.b_2b_3\ldots$. Everything of 10.3.7–10.3.8 carries over with $10 \to 2$: *fixed points:* 0 (and $1 \equiv 0$) only ($x =$

$k/(2-1)$); *periodic points:* $x = k/(2^p-1)$, e.g. $x = 1/(2^p-1) = 0.\overline{0\cdots01}_2$ has period p, with multiplier $2^p > 1$ (all unstable); *aperiodic orbits:* any irrational x_0; *sensitivity:* two points differing at bit m separate by a factor 2 each step, Liapunov exponent $\ln 2$; *dense orbit:* concatenate all finite bit strings, $x_0 = 0.\,0\,1\;00\,01\,10\,11\;000\,001\cdots$, so every finite bit pattern of every length appears. ■

10.3.10 *(Exact solutions of the logistic map at $r = 4$)*

Solution. **(a)** Let $\theta_{n+1} = 2\theta_n \bmod 1$ and $x_n = \sin^2(\pi\theta_n)$. Since $\sin^2(\pi\theta)$ has period 1 in θ, the "mod 1" is harmless:

$$x_{n+1} = \sin^2(2\pi\theta_n) = 4\sin^2(\pi\theta_n)\cos^2(\pi\theta_n) = 4x_n(1-x_n),$$

for any θ_0. So every orbit of the doubling map yields an exact orbit of the logistic map with $r = 4$.
(b) Examples ($\theta_0 = \frac{1}{3}$: θ_n alternates $\frac{1}{3}, \frac{2}{3}$, so $x_n \equiv \frac{3}{4}$, the fixed point $1 - 1/r$; $\theta_0 = \frac{1}{7}$: $\frac{1}{7}, \frac{2}{7}, \frac{4}{7}$ gives a 3-cycle; $\theta_0 = \sqrt{2} - 1$: aperiodic, irregular): Every orbit type of the shift (fixed, periodic of every period, aperiodic, dense) occurs for the logistic map at $r = 4$, and the exponential sensitivity is $\lambda = \ln 2$. ■

10.3.11 *(Subcritical flip)*

$f(x) = -(1+r)x - x^2 - 2x^3$.

Solution. **(a)** $f'(0) = -(1+r)$, so $x^* = 0$ is stable for $-2 < r < 0$ and unstable for $r > 0$ (also for $r < -2$).
(b) At $r = 0$, $f'(0) = -1$: the multiplier passes through -1, a flip bifurcation.
(c) Let $a = -(1+r)$ so $f(x) = ax - x^2 - 2x^3$. Expanding,

$$f^2(x) = a^2x - a(1+a)x^2 - 2a^3x^3 + O(x^4), \qquad -a(1+a) = -r(1+r).$$

At $r = 0$ this is $f^2(x) = x + 2x^3 + O(x^4)$, so $x_{n+2} - x_n \approx 2x_n^3$: the fixed point repels, consistent with a subcritical bifurcation. For small r the 2-cycle satisfies $f^2(x) = x$, i.e. (dividing by x)

$$(2r + r^2) - r(1+r)x + 2(1+r)^3x^2 + O(x^3) = 0 \implies x^2 \approx -r.$$

So there is a 2-cycle $x \approx \pm\sqrt{-r}$ for $r < 0$ (numerically, at $r = -0.1$ the cycle point is 0.250 against the leading-order estimate $\sqrt{0.1} = 0.316$; at $r = -0.01$ it is 0.094 against 0.1). Its multiplier is $(f^2)'(x) \approx 1 + 2r + 6x^2 \approx 1 + 2r - 6r = 1 - 4r > 1$ for $r < 0$: it is *unstable*. As $r \to 0^-$ the cycle shrinks onto the origin, which is the signature of a subcritical flip: an unstable 2-cycle surrounds the stable fixed point for $r < 0$ and coalesces with it at $r = 0$.
(d) For $r < 0$ (origin stable): orbits that start inside the unstable 2-cycle ($|x_0| \lesssim \sqrt{-r}$) converge to 0; those outside are repelled. For $r > 0$ the origin is unstable and, since $f(x) \sim -2x^3$ for large $|x|$ and I found no other attractor, orbits escape: numerically, with $r = 0.1$ every start I tried (from $x_0 = 0.01$ to 0.5) diverged to $\pm\infty$, and with $r = -0.1$ the starts $0.01, 0.1, 0.2$ converged to 0 while 0.3 and 0.5 diverged (consistent with a cycle at ≈ 0.25). So for $r > 0$ there is no bounded attractor near 0: orbits starting near 0 move away and diverge. ■

10.3.12 *(Numerics of superstable cycles)*

Solution. **(a)** A cycle is superstable iff it contains the critical point $x = \frac{1}{2}$. Hence R_n is defined implicitly by

$$f^{2^n}\left(\tfrac{1}{2}, R_n\right) = \tfrac{1}{2}, \qquad f(x,r) = rx(1-x),$$

with R_n the *smallest* root for which $\frac{1}{2}$ has exact period 2^n.
(b) Solving this with 30-digit Newton/secant iteration (starting from $R_{n-1}+(R_{n-1}-R_{n-2})/4.669$):

n	R_n	n	R_n
0	2	4	3.5666674
1	3.2360680	5	3.5692435
2	3.4985617	6	3.5697953
3	3.5546409	7	3.5699135

To five significant figures: $R_2 = 3.4986$, $R_3 = 3.5546$, $R_4 = 3.5667$, $R_5 = 3.5692$, $R_6 = 3.5698$, $R_7 = 3.5699$.
(c) $\dfrac{R_6 - R_5}{R_7 - R_6} = \dfrac{0.00055178}{0.00011817} \approx 4.669$ (more precisely 4.66916), agreeing with Feigenbaum's $\delta = 4.6692\ldots$ to four digits. (The earlier ratios are $4.681, 4.663, 4.668, 4.669, 4.669$ for $n = 3, \ldots, 7$.) ■

10.3.13 *(Tantalizing patterns)*

Solution. **(a)** The dark smooth tracks are the curves $r \mapsto f^k(x_m, r)$ for $k = 1, 2, 3, \ldots$, i.e. the first few iterates of the critical point: $x = \frac{r}{4}$ (the upper boundary of the attractor), $x = \frac{r^2}{4}\big(1 - \frac{r}{4}\big)$ (the lower boundary), and so on. They are dark because orbit points pile up there: near x_m the map is a fold, $f(x) \approx f(x_m) - \frac{r}{2}(x - x_m)^2$, and a fold turns a spread-out set of points into a set squeezed against the maximum value, giving a square-root singularity in the density of the invariant measure at every image $f^k(x_m)$.
(b) The corner of the wedge is where these tracks meet at the unstable fixed point $x^* = 1 - \frac{1}{r}$. Since $f(x) = f(1-x)$, $f^3(x_m) = x^*$ holds iff $f^2(x_m) \in \{x^*,\ 1 - x^* = \frac{1}{r}\}$; taking $f^2(x_m) = \frac{1}{r}$:

$$\frac{r^2(4-r)}{16} = \frac{1}{r} \iff r^4 - 4r^3 + 16 = 0 \implies \boxed{r_c \approx 3.678573510}.$$

I checked numerically that $f^3(\frac{1}{2}, r_c) = 1 - 1/r_c = 0.72815549\ldots$ to 30 digits. This is the value at which the two chaotic bands merge into one (all tracks $f^k(x_m)$, $k \geq 3$, pass through x^* there). ■

4 Periodic Windows

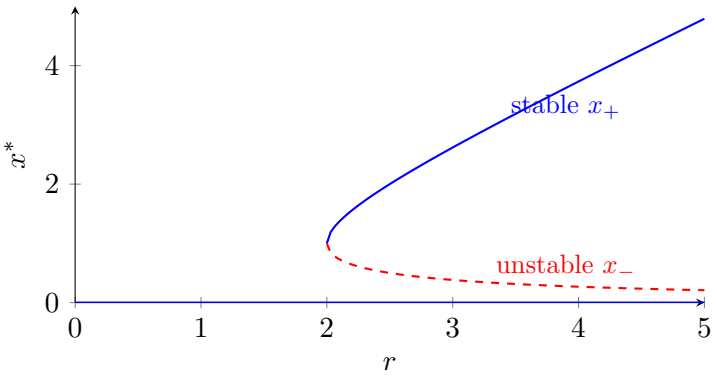
10.4.1 *(Exponential map $x_{n+1} = re^{x_n}$, $r > 0$)*

Solution. **(a)** Fixed points solve $re^x = x$. Let $h(x) = re^x - x$; $h' = re^x - 1 = 0$ at $x = -\ln r$, where $h = 1 + \ln r$. So fixed points exist iff $1 + \ln r \leq 0$, i.e. $r \leq 1/e$: two of them for $r < 1/e$ (cobweb: the curve re^x crosses the diagonal twice), none for $r > 1/e$. With $f' = re^x = x^*$ at a fixed point, the smaller one has $f' < 1$ (stable) and the larger has $f' > 1$ (unstable). Orbits starting below the unstable fixed point converge to the stable one; those above it diverge to $+\infty$. For $r > 1/e$ the curve lies above the diagonal and every orbit increases to $+\infty$.
(b) At a tangency $f(x) = x$ and $f'(x) = 1$, so $x^* = 1$ and $r = 1/e$. Check: $f'' = re^x = 1/e \cdot e = 1 \neq 0$ and $\partial f/\partial r = e^x \neq 0$: a saddle-node (**tangent bifurcation**) at $r = 1/e$.
(c) Just below $r = 1/e$ ($r = 0.36$, $x_0 = 0$) the orbit rises and creeps up very slowly toward the stable fixed point (after 300 iterations it is 0.806 and still approaching ≈ 0.80), because it passes through the near-tangency. Just above ($r = 0.38$) the orbit also creeps through the "bottleneck"

near $x = 1$ for many iterations and then escapes and blows up (here $x_n > 5$ after 23 steps; the bottleneck takes of order $\pi/\sqrt{\cdot}$ steps as $r \to 1/e$, cf. 10.7.8). ■

10.4.2 *($x_{n+1} = rx_n^2/(1+x_n^2)$)*

Solution. Fixed points: $x = 0$ (all r) and $x^2 - rx + 1 = 0$, i.e. $x_\pm = \frac{r \pm \sqrt{r^2-4}}{2}$, existing for $r \geq 2$. $f'(x) = \dfrac{2rx}{(1+x^2)^2}$, so $f'(0) = 0$ (superstable for all r), and at $x_\pm$, using $1 + x^2 = rx$: $f' = \dfrac{2}{rx_\pm}$. For $r > 2$: $x_+ > 1$ so $rx_+ > 2$ and $0 < f'(x_+) < 1$ (stable); $x_- < 1$, $rx_- = 1 + x_-^2 \in (1, 2)$ so $f'(x_-) > 1$ (unstable). At $r = 2$: $x_\pm = 1$, $f' = 1$, $f'' \neq 0$: a **saddle-node bifurcation** at $r = 2$ creating the stable x_+ and the unstable x_-.



(Solid = stable, dashed = unstable.) For $r > 2$ the system is bistable, x_- being the basin boundary. *Intermittency?* No. f is increasing on $x \geq 0$ (and $f \geq 0$ everywhere), so orbits are monotone and converge to a fixed point; there are no cycles and no chaos. A ghost bottleneck near $x = 1$ for r slightly below 2 makes some orbits take long to pass, but after the bottleneck they fall into the stable fixed point 0 and are never re-injected, so no intermittent bursting occurs. ■

10.4.3 *(A superstable 3-cycle)*

$x_{n+1} = 1 - rx_n^2$.

Solution. Superstable means the cycle contains the critical point $x = 0$: $0 \to f(0) = 1 \to f(1) = 1 - r \to f(1-r) = 1 - r(1-r)^2 = 0$. Hence $r(1-r)^2 = 1$:

$$\boxed{r^3 - 2r^2 + r - 1 = 0}, \qquad r \approx 1.7549$$

(the real root; the other two roots are complex). In the logistic parametrization this corresponds to the superstable 3-cycle at $r \approx 3.8319$ (10.4.4). ■

10.4.4 *(Superstable 3-cycle of the logistic map)*

Solution. Solve $f^3(\frac{1}{2}, r) = \frac{1}{2}$ for r near 3.83 (Newton/bisection on $g(r) = f^3(\frac{1}{2}, r) - \frac{1}{2}$):

$$\boxed{r \approx 3.83187406} \quad (3.8318740553\ldots).$$

(Equivalently, the root 1.75488 of 10.4.3 transformed by the conjugacy of 10.3.5.) ■

10.4.5 *(Band merging and crisis)*

Solution. I zoomed the orbit diagram into $3.7 \le r \le 3.9$ (Fig. below; lines at 3.8284, 3.8495, 3.857). The stable 3-cycle appears at $r = 1 + \sqrt{8} = 3.8284$ (10.4.10). Its first period doubling is at $r = 3.84147$ (multiplier $= -1$, computed). Counting the distinct points visited by the attractor: 3 at $r = 3.84$, 6 at 3.844–3.846, 12 at 3.848–3.849, 48 at 3.8494, and an aperiodic orbit at 3.8496: the doublings accumulate near $r \approx 3.8495$, after which there are three small chaotic bands (the orbit visits them cyclically, so each band maps to the next). For slightly larger r (about 3.857, read from the figure) the three bands merge and the attractor suddenly jumps in size to a much larger interval that nearly fills $[0.15, 0.97]$. This discontinuous change in the size of the attractor is a crisis. (I read the merging value off the diagram; I did not locate it to more than three digits.) ■

10.4.6 *(A superstable cycle)*

Solution. With $r = 3.7389149$ and $x_0 = \frac{1}{2}$ the iterates are 0.5, 0.934729, 0.228115, 0.658342, 0.840986, 0.500000, The orbit returns to $\frac{1}{2}$ after 5 steps: a superstable cycle of period $\boxed{5}$. (Refining, $r = 3.738914913$, with $f^5(\frac{1}{2}) - \frac{1}{2} \approx 10^{-15}$.) ■

10.4.7 *(Iteration patterns)*

Solution. **(a)** $x_1 = f(\frac{1}{2}) = \frac{r}{4} > \frac{1}{2}$ for $r > 2$, so the first letter is R. $x_2 = f(x_1) = \dfrac{r^2(4-r)}{16}$ is to the left of $\frac{1}{2}$ iff $r^2(4-r) < 8 \iff r^3 - 4r^2 + 8 > 0 \iff (r-2)(r^2 - 2r - 4) > 0$, and for $r > 2$ this holds iff $r > 1 + \sqrt{5}$ (the superstable 2-cycle of 10.3.2). Hence for $r > 1 + \sqrt{5}$ the first two letters are $\boxed{RL}$.
(b) For the cycle of 10.4.6: 0.9347 (R), 0.2281 (L), 0.6583 (R), 0.8410 (R), then 0.5: the pattern is $\boxed{RLRR}$. ■

10.4.8 *(Intermittency in the Lorenz equations)*

$\sigma = 10$, $b = 8/3$, r near 166.

Solution. **(a)** At $r = 166$, integrating from $(1, 2, 3)$ (the only start I used), the trajectory settles onto a stable limit cycle: its z-maxima repeat exactly with two values per period (196.1 and 227.5), and the xz projection is a closed curve. **(b)** At $r = 166.2$ the maxima of z stay on the two limit-cycle values for long stretches (the trajectory looks like the old cycle), interrupted by irregular chaotic bursts (right panel): intermittency.
(c) I classified a maximum as "burst" if it deviates by more than 3 from the two cycle values, and measured over $200 < t < 1500$:

r	166.1	166.2	166.4	166.7	167.2
fraction of time in bursts	0.16	0.22	0.35	0.45	0.58
burst episodes in 1300 time units	302	474	634	792	822

So bursts occupy an increasing share of the time and occur more often as r increases; their mean duration (fraction $\times 1300$ / episodes) rises from about 0.7 to about 0.9 time units, i.e. the bursts last longer. (This is a crude measure with a somewhat arbitrary threshold, so the trend, not the exact numbers, is the result.) ■

10.4.9 *(Period doubling in the Lorenz equations)*

Solution. Discarding the transient ($t < 600$) and plotting the xy projection: At $r = 148.5$ the attractor is a single closed loop: a stable limit cycle. At $r = 147.5$ the loop has doubled: the trajectory goes around twice, with two nearly parallel strands, before closing; this is the period-doubled version of the cycle (the first doubling in the cascade into chaos as r decreases, cf. 9.5.3). ■

10.4.10 *(The birth of period 3)*

Solution. Use the conjugate map $x_{n+1} = x_n^2 + k$ ($k = r(2-r)/4$ by 10.3.5; the maximum is at $x = 0$). Let the 3-cycle be $a \to b \to c \to a$ with $b = a^2 + k$, $c = b^2 + k$, $a = c^2 + k$. A tangent bifurcation means the multiplier is $+1$: $8abc = 1$. Let $s_1 = a+b+c$, $s_2 = ab+bc+ca$, $s_3 = abc = \frac{1}{8}$.
Differences: $b - c = (a - b)(a + b)$ and cyclically; multiplying the three relations and cancelling $(a-b)(b-c)(c-a) \neq 0$ gives $(a+b)(b+c)(c+a) = 1$, i.e. $s_1 s_2 - s_3 = 1$, so $s_1 s_2 = \frac{9}{8}$.
Sum: $s_1 = \sum a^2 + 3k = s_1^2 - 2s_2 + 3k$, so $3k = s_1 - s_1^2 + 2s_2$.
Products: $s_2 = \sum ab = \sum a(a^2 + k) = \sum a^3 + ks_1 = s_1^3 - 3s_1 s_2 + 3s_3 + ks_1$.
Eliminating $s_2 = \frac{9}{8s_1}$ and k gives the single equation

$$16s_1^4 + 8s_1^3 - 54s_1 - 27 = 0 \iff (2s_1 - 3)(2s_1 + 1)(4s_1^2 + 6s_1 + 9) = 0.$$

The factor $4s_1^2 + 6s_1 + 9$ has no real roots. The root $s_1 = \frac{3}{2}$ gives $k = \frac{1}{4}$: the trivial triple fixed point $a = b = c = \frac{1}{2}$. The root $s_1 = -\frac{1}{2}$ gives $s_2 = -\frac{9}{4}$ and

$$k = \tfrac{1}{3}\big(s_1 - s_1^2 + 2s_2\big) = \tfrac{1}{3}\big(-\tfrac{1}{2} - \tfrac{1}{4} - \tfrac{9}{2}\big) = -\tfrac{7}{4}.$$

Then $k = \dfrac{r(2-r)}{4} = -\dfrac{7}{4} \Rightarrow r^2 - 2r - 7 = 0 \Rightarrow$

$$\boxed{r = 1 + \sqrt{8} \approx 3.8284.}$$

■

10.4.11 *(Repeated exponentiation)*

$x_1 = a$, $x_{n+1} = a^{x_n}$.

Solution. This is the iteration of $f(x) = a^x$ from $x_1 = a$. Fixed points solve $a^x = x$, with $f'(x^*) = x^* \ln a$.
Case $a > 1$. f is increasing, so the sequence is monotone ($x_2 = a^a > a = x_1$). It converges iff a fixed point exists above x_1, i.e. iff $a^x = x$ has a solution, which happens iff $a \leq e^{1/e} \approx 1.4447$ (tangency at $x = e$: $a^e = e$ and $e \ln a = 1$). For $1 < a \leq e^{1/e}$ the sequence increases to the smaller fixed point x^* ($x^* = e$ when $a = e^{1/e}$; numerically $a = 1.2 \to 1.2577$, $1.4 \to 1.887$, $1.4446 \to 2.675$); for $a > e^{1/e}$ it diverges to $+\infty$ (e.g. $a = 1.45, 2$).
Case $a = 1$. $x_n \equiv 1$.
Case $0 < a < 1$. f is decreasing, with a unique fixed point $x^* = W(|\ln a|)/|\ln a|$ and $f'(x^*) = -W(|\ln a|)$. Hence it is stable iff $W(|\ln a|) < 1 \iff |\ln a| < e \iff a > e^{-e} \approx 0.0660$, and the sequence converges (oscillating) to x^* (e.g. $a = 0.5 \to 0.641$, $0.1 \to 0.399$). At $a = e^{-e}$ there is a flip bifurcation, and for $0 < a < e^{-e}$ the fixed point is unstable and the sequence settles onto a **2-cycle**: the odd and even terms converge to different limits (numerically $a = 0.05$: 0.137, 0.663; $a = 0.01$: 0.0131, 0.9415; as $a \to 0$ the limits approach 0 and 1). Since f is decreasing, f^2 is increasing, so cycles of period > 2 (and chaos) are impossible: for every $a > 0$ the long-term behavior is a fixed point, a 2-cycle, or divergence. So the "more complicated" behavior is just the 2-cycle for $a < e^{-e}$ (the sequence has two limit points rather than one); I did not find anything more complex. ■

5 Liapunov Exponent

10.5.1 *(Linear map $x_{n+1} = rx_n$)*

Solution. $\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln|f'(x_i)| = \ln|r|$ since $f' \equiv r$. (Negative for $|r| < 1$, where $x_n \to 0$; positive for $|r| > 1$.) ■

10.5.2 *(Decimal shift map)*

Solution. $f' \equiv 10$ wherever it is defined, so $\lambda = \ln 10 \approx 2.30 > 0$ for every orbit: each decimal digit is lost per step. ■

10.5.3 *(Tent map for $r \leq 1$)*

$x_{n+1} = T_r(x_n) = r\min(x_n, 1 - x_n)$, $0 \leq x \leq 1$.

Solution. For $0 \leq r < 1$: $T_r(x) \leq rx$ for all x (for $x \geq \frac{1}{2}$, $r(1-x) \leq \frac{r}{2} \leq rx$), so $x_n \leq r^n x_0 \to 0$: $x^* = 0$ is globally stable and $\lambda = \ln r < 0$. At $r = 1$: $T_1(x) = \min(x, 1-x)$ maps $[\frac{1}{2}, 1]$ onto $[0, \frac{1}{2}]$ and fixes every point of $[0, \frac{1}{2}]$; so after at most one step every orbit lands on a fixed point in $[0, \frac{1}{2}]$: a continuum of neutrally stable fixed points, $\lambda = \ln 1 = 0$. ■

10.5.4 *(No windows for the tent map)*

Solution. For $1 < r \leq 2$, $|T_r'| = r > 1$ wherever differentiable. A periodic orbit of period n that avoids the kink has multiplier $(\pm r)^n$ of modulus $r^n > 1$, so *every* periodic orbit is unstable. A periodic window of the logistic type is a parameter interval with a *stable* periodic orbit, so there are none. Equivalently, the Liapunov exponent is $\lambda = \ln r > 0$ for every orbit not hitting the kink, for every $r > 1$: it never dips below zero, whereas the logistic map has $\lambda < 0$ inside each window. (The sine/logistic maps differ because f' is small near the maximum, which allows the superexponential contraction needed for stable cycles.) ■

10.5.5 *(Orbit diagram of the tent map)*

Solution. See the last panel of the multi-panel orbit diagram above (10.2.4–10.2.8). For $r < 1$ the orbit goes to 0; for $1 < r < \sqrt{2}$ the attractor consists of two bands that are visited alternately; they merge at $r = \sqrt{2}$; for $\sqrt{2} < r \leq 2$ a single band is filled, whose upper edge is $r/2$ and lower edge $r(1 - r/2)$ (the dark tracks), and there are no windows – the picture is a smooth widening wedge. (At exactly $r = 1$ the plot shows a spurious point because of the continuum of fixed points.) ■

10.5.6 *(Liapunov exponent of the sine map)*

$x_{n+1} = r\sin \pi x_n$.

Solution. Computed as $\lambda(r) = \frac{1}{N} \sum \ln|r\pi \cos \pi x_n|$ with $N = 5000$ after 500 transient steps, on 1400 values of $r \in [0.3, 1]$: $\lambda < 0$ for the stable fixed point and cycles, touching 0 at each bifurcation (10.5.7); λ first becomes positive at $r \approx 0.8659$ (the onset of chaos, consistent with $r_\infty \approx 0.8656$), and then shows downward spikes into $\lambda < 0$ at the periodic windows; $\lambda \approx 0.69$ at $r = 1$ (the largest value in $[0.3, 1]$). ■

10.5.7 *($\lambda = 0$ at period-doubling)*

Solution. For a stable cycle $\{x_1, \ldots, x_p\}$ the orbit visits these points periodically, so

$$\lambda = \frac{1}{p}\sum_{i=1}^{p} \ln|f'(x_i)| = \frac{1}{p}\ln\left|(f^p)'(x_1)\right| = \frac{1}{p}\ln|\mu|,$$

where μ is the multiplier. At the period-doubling bifurcation value r_n the 2^{n-1}-cycle has $\mu = -1$, so $|\mu| = 1$ and $\lambda = \frac{1}{2^{n-1}}\ln 1 = 0$. By continuity of $\lambda(r)$, λ equals 0 at each r_n (the dips of λ to 0 in the figure of 10.5.6). ■

6 Universality and Experiments

10.6.1 *(Naive approach (sine map $x_{n+1} = r\sin\pi x_n$))*

Solution. **(a)** Orbit diagram from 200 equally spaced $r \in (0, 1]$, plotting $x_{700}, \ldots, x_{1000}$ from a random start: It shows $x^* = 0$ for $r < \frac{1}{\pi}$, a stable fixed point, then a period-doubling cascade starting near $r \approx 0.72$, chaos beyond $r \approx 0.866$, and periodic windows.
(b) Instead of locating r_n by watching the orbit (slow near each bifurcation: critical slowing down), I solved directly for the flip: for the 2^{n-1}-cycle with a point x near $\frac{1}{2}$, solve $f^N(x) = x$ and $(f^N)'(x) = -1$ ($N = 2^{n-1}$) simultaneously (30-digit Newton), giving

n	1	2	3	4	5	6
r_n	0.71996	0.83327	0.85861	0.86408	0.86526	0.86551

For comparison, the naive method (detect the first r on a grid of step 10^{-5} where the orbit looks period 2^n after 3000 iterations) gives 0.71915, 0.83295, 0.85849, 0.86403, 0.86525, 0.86551: too low by 8×10^{-4}, 3×10^{-4}, 1×10^{-4}, …, because just below r_n the approach to the old cycle is so slow that an orbit that has not yet converged is mistaken for the new cycle.
(c) $\dfrac{r_n - r_{n-1}}{r_{n+1} - r_n}$: 4.471 ($n = 2$), 4.629, 4.661, 4.667 ($n = 5$), approaching $\delta = 4.669$. ■

10.6.2 *(Superstable cycles to the rescue)*

Solution. **(a)** The superstable value R_n is defined by a *sign-changing equation* $f^{2^n}(\frac{1}{2}, R_n) = \frac{1}{2}$ with a nonzero slope (a simple root), so standard root-finding converges quadratically; there is no need to wait for any transient to converge, since the iterates are computed explicitly from $x = \frac{1}{2}$. By contrast r_n is defined by the orbit becoming marginal, where the convergence of the orbit itself is infinitely slow.
(b) Solving $f^{2^n}(\frac{1}{2}, R_n) = \frac{1}{2}$ with $f = r\sin\pi x$: $R_0 = 0.5$, and

n	1	2	3	4	5	6
R_n	0.777734	0.846382	0.861450	0.864694	0.865390	0.865539

($R_7 = 0.865571$). The ratios $\frac{R_{n-1}-R_{n-2}}{R_n-R_{n-1}}$ for $n = 2, \ldots, 7$ are 4.046, 4.556, 4.645, 4.664, 4.668, 4.669, so $\delta \approx 4.669$ (the best estimate), with errors of order 10^{-3} at $n = 5$ – much better than the naive approach. ■

10.6.3 *(Qualitative universality of patterns)*

Solution. **(a)** Cobwebs for the superstable 6-cycles (starting from $x_0 = \frac{1}{2}$): **(b)** Iterating from $x_0 = \frac{1}{2}$ and recording the side of $\frac{1}{2}$ on which each of $x_1, \ldots, x_5$ falls, both cycles have iteration pattern $\boxed{RLRRR}$. The patterns coincide, as the U-sequence predicts. ■

10.6.4 *(Period 4)*

Solution. **(a)** A superstable cycle of period n has $n-1$ letters (for $x_1, \ldots, x_{n-1}$). The first letter is R (the image of the maximum point is the maximum value, to its right). For $n \geq 3$ the second letter must be L: otherwise $x_1, x_2 > x_m$ with $x_2 = f(x_1) < x_1$; f is decreasing on the right of x_m, so $x_3 = f(x_2) \in (x_2, x_1)$, again to the right of x_m, and inductively every iterate stays in $(x_m, x_1]$, so the orbit never returns to x_m, contradicting that the cycle contains x_m. So the pattern begins RL, and for $n = 4$ the third letter is R or L: only RLR and RLL are possible.
(b) Numerically ($f = rx(1-x)$): the cycle with pattern RLR occurs at $r = 3.4985617$ (the period doubling of the superstable 2-cycle, the second entry $2 \cdot 2$ of the U-sequence, which begins $1, 2, 4, 6, 5, 3, \ldots$) and the cycle with pattern RLL at $r = 3.9602701$ (the later entry "4" of the U-sequence $\ldots, 5, 6, \mathbf{4}, 6, 5, 6$ near the end of the diagram). So RLL occurs at larger r than RLR. (I verified the two values and patterns by solving $f^4(\frac{1}{2}, r) = \frac{1}{2}$; I have not given an independent proof of the ordering beyond the U-sequence theorem.) ■

10.6.5 *(Unfamiliar later cycles)*

Solution. **(a)** Their iteration patterns (letters for $x_1, x_2, \ldots$) are $RLLR$, $RLLRR$, RLL, $RLLLR$, $RLLL$, $RLLLL$.
(b) Yes, they are hard to find, mainly because their windows are extremely narrow. Scanning r in fine steps around each value I found widths of roughly 5×10^{-4}, 7×10^{-5}, 6×10^{-4}, 2×10^{-5}, 3×10^{-5}, 2×10^{-6} for the periods $5, 6, 4, 6, 5, 6$. A coarse grid in r steps right over them, and an orbit plotted at a value of r that is only slightly off the window is simply chaotic (the surrounding chaotic attractor nearly fills the interval there). So the parameter must be specified to about the width of the window (here up to 7 digits), and near the edges of a window a long transient is needed before the cycle is reached. ■

10.6.6 *(Hao and Zheng's trick)*

Solution. **(a)** For $f(x) = r - x^2$ the maximum is at $x = 0$. For the pattern $RLLR$ the superstable cycle is $0 \to x_1 \to x_2 \to x_3 \to x_4 \to 0$, with $x_1, x_4 > 0$ and $x_2, x_3 < 0$. Run it *backwards*: $x_5 = 0 = r - x_4^2$ with $x_4 > 0 \Rightarrow x_4 = \sqrt{r} = R(0)$; $x_3 < 0$ with $x_4 = r - x_3^2 \Rightarrow x_3 = -\sqrt{r - x_4} = L(x_4)$; $x_2 = L(x_3)$; $x_1 = R(x_2)$; and finally $x_1 = f(0) = r$. Hence $r = R(L(L(R(0))))$, which written out is

$$r = \sqrt{r + \sqrt{r + \sqrt{r - \sqrt{r}}}}\,.$$

(Using $x_4 = \sqrt{r}$, $x_3 = -\sqrt{r - \sqrt{r}}$, $x_2 = -\sqrt{r + \sqrt{r - \sqrt{r}}}$, $x_1 = \sqrt{r - x_2} = \sqrt{r + \sqrt{r + \sqrt{r - \sqrt{r}}}}$.)

(b) Iterating $r_{n+1} = \sqrt{r_n + \sqrt{r_n + \sqrt{r_n - \sqrt{r_n}}}}$ from $r_0 = 2$: 1.91388, 1.88132, 1.86877, 1.86390, 1.86200, 1.86126, and after 40 iterations $r = 1.860782522$ (the error shrinks by a factor of about 0.4 per iteration: geometric convergence, observed numerically).

(c) With $r = 1.860782522$ and $x_0 = 0$: $x_1 = 1.860783 > 0$ (R), $x_2 = -1.601729$ (L), $x_3 = -0.704754$ (L), $x_4 = 1.364105$ (R), $x_5 = r - x_4^2 \approx 0$. So the orbit closes after five steps with pattern RLLR, as desired. ■

7 Renormalization

10.7.1 *(Hands-on with the functional equation)*

$g(x) = \alpha g^2(x/\alpha)$.

Solution. **(a)** Take $g(x) \approx 1 + c_2 x^2$. Then $g^2(y) = g(g(y)) = 1 + c_2(1 + c_2 y^2)^2 = (1 + c_2) + 2c_2^2 y^2 + O(y^4)$, so

$$\alpha g^2(x/\alpha) = \alpha(1 + c_2) + \frac{2c_2^2}{\alpha} x^2 + O(x^4).$$

Equating with $g(x) = 1 + c_2 x^2$: $\alpha(1 + c_2) = 1$ and $c_2 = 2c_2^2/\alpha$, i.e. $\alpha = 2c_2$. Then $2c_2(1 + c_2) = 1 \Rightarrow 2c_2^2 + 2c_2 - 1 = 0 \Rightarrow c_2 = \frac{-1 \pm \sqrt{3}}{2}$. For a maximum at 0 we need $c_2 < 0$:

$$\boxed{c_2 = -\tfrac{1+\sqrt{3}}{2} \approx -1.366, \qquad \alpha = 2c_2 = -(1 + \sqrt{3}) \approx -2.732.}$$

(b) With $g \approx 1 + c_2 x^2 + c_4 x^4$ the same expansion gives: constant term $\alpha(1 + c_2 + c_4) = 1$; x^2: $\alpha = 2c_2 + 4c_4$; x^4: $c_4 \alpha^3 = (2c_2 + 4c_4)c_4 + (c_2 + 6c_4)c_2^2$. Solving numerically (fsolve, residual $< 10^{-12}$):

$$\alpha \approx -2.534, \qquad c_2 \approx -1.522, \qquad c_4 \approx 0.128,$$

compared with the "exact" -2.5029, -1.527, 0.1048. The two-term fit improves α from -2.73 (error 9%) to -2.53 (error 1%). ■

10.7.2 *(Rescaling and inversion)*

Solution. Let $y_{n+1} = f(y_n)$ and set $x_n = \alpha y_n$. Then $x_{n+1} = \alpha y_{n+1} = \alpha f(y_n) = \alpha f(x_n/\alpha)$: the rescaled map is $\tilde{f}(x) = \alpha f(x/\alpha)$. Its second iterate is $\tilde{f}^2(x) = \alpha f\big(\tilde{f}(x)/\alpha\big) = \alpha f\big(f(x/\alpha)\big) = \alpha f^2(x/\alpha)$. Taking $f = f(\cdot, R_1)$ gives the claim that rescaling ($|\alpha|$) and inversion ($\alpha < 0$ flips $x \to -x$) converts $f^2(x, R_1)$ into $\alpha f^2(x/\alpha, R_1)$; the sign of α is what restores the orientation of the maximum (the second iterate has a minimum at 0 where f has a maximum). ■

10.7.3 *(Scale freedom of g)*

Solution. Let $h(x) = \mu g(x/\mu)$. Then

$$\alpha h^2(x/\alpha) = \alpha\mu\, g\Big(\tfrac{1}{\mu} \cdot \mu g\big(\tfrac{x}{\alpha\mu}\big)\Big) = \mu\big[\alpha g^2\big(\tfrac{x/\mu}{\alpha}\big)\big] = \mu\, g(x/\mu) = h(x),$$

using $g(z) = \alpha g^2(z/\alpha)$ with $z = x/\mu$. So h solves the same equation with the same α; the scale μ is fixed only by a normalization such as $g(0) = 1$. ■

10.7.4 *(Wildness of g)*

Solution. Suppose $g(x^*) = x^*$. Then $g(\alpha x^*) = \alpha g^2(x^*) = \alpha g(g(x^*)) = \alpha g(x^*) = \alpha x^*$, so αx^* is also a fixed point, and iterating, so is $\alpha^k x^*$ for all $k \geq 1$. Does a fixed point $x^* \neq 0$ exist?

$g(0) = 1 > 0$ is the maximum of g (normalized $g(0) = 1$, $g \le 1$), so $g(x) < x$ for $x > 1$ and $g(x) - x$ changes sign on $(0, 1]$: there is a fixed point $x^* \in (0, 1]$. Since $|\alpha| > 1$ the points $\alpha^k x^*$ go off to infinity with alternating signs (as $\alpha < 0$), giving infinitely many crossings of $y = x$. Because g is even, $g(-x^*) = x^*$, so $-\alpha^k x^*$ are points where $g(x) = -x$: infinitely many crossings of $y = -x$ as well. So g crosses both lines $y = \pm x$ infinitely often, and therefore has infinitely many wiggles as x ranges over $\mathbb{R}$. ■

10.7.5 *(Crudest estimate of α)*

$f(x, r) = r - x^2$.

Solution. **(a)** $R_0 = 0$ (superstable fixed point: $f(0, r) = r = 0$) and $R_1 = 1$ (superstable 2-cycle: $f^2(0) = r - r^2 = 0$). Hence $f(x, R_0) = -x^2$, and $f(x, R_1) = 1 - x^2$, $f^2(x, R_1) = 1 - (1 - x^2)^2 = 2x^2 - x^4$, so

$$\alpha f^2(x/\alpha, R_1) = \alpha\Big(\frac{2x^2}{\alpha^2} - \frac{x^4}{\alpha^4}\Big) = \frac{2}{\alpha}x^2 - \frac{x^4}{\alpha^3}.$$

(b) Matching the $O(x^2)$ coefficients with $f(x, R_0) = -x^2$: $\dfrac{2}{\alpha} = -1 \Rightarrow \alpha = -2$. ■

10.7.6 *(Improved estimate of α)*

Solution. Compare $\alpha f^2(x/\alpha, R_1) = \frac{2}{\alpha}x^2 + O(x^4)$ with $\alpha^2 f^4(x/\alpha^2, R_2)$, where R_2 is the superstable 4-cycle ($f^4(0, R_2) = 0$). Near 0, $f^4(y, R_2) = Ay^2 + O(y^4)$. Differentiating the chain of compositions (with $R = R_2$, $c_3 = f^3(0) = R - (R - R^2)^2$):

$$f^4(y) \approx R - c_3^2 + 8R(R - R^2)c_3\, y^2, \qquad f^4(0) = 0 \Rightarrow c_3^2 = R,$$

so $A = 8R(R - R^2)c_3$. Then $\alpha^2 f^4(x/\alpha^2, R_2) = \alpha^2 A\, x^2/\alpha^4 = \dfrac{A}{\alpha^2}x^2$. Matching with $\frac{2}{\alpha}x^2$ gives $\alpha = A/2$. With $R_2 = 1.3107026$, $c_3 = 1.1449$, $A = -4.889$:

$$\boxed{\alpha \approx -2.444}$$

(compared with -2.5029: error 2.3%; the crude estimate -2 had 20%). ■

10.7.7 *(Quartic maxima)*

$f(x, r) = r - x^4$.

Solution. *Method of 10.7.1.* Take $g(x) \approx 1 + c_4 x^4$. Then $g^2(y) = 1 + c_4(1 + c_4 y^4)^4 \approx (1 + c_4) + 4c_4^2 y^4$, so $\alpha g^2(x/\alpha) = \alpha(1 + c_4) + 4c_4^2 x^4/\alpha^3$. Matching: $\alpha(1 + c_4) = 1$ and $c_4 = 4c_4^2/\alpha^3 \Rightarrow c_4 = \alpha^3/4$. Hence $\alpha + \alpha^4/4 = 1$, i.e. $\alpha^4 + 4\alpha - 4 = 0$, whose negative root is $\boxed{\alpha \approx -1.835}$ (then $c_4 \approx -1.545$).
Method of 10.7.5. $R_0 = 0$, $R_1 = 1$ again (since $r - r^4 = 0$ at $r = 1$). $f(x, R_0) = -x^4$ and $\alpha f^2(x/\alpha, R_1) = \alpha(1 - (1 - x^4/\alpha^4)^4) \approx 4x^4/\alpha^3$. Matching $-1 = 4/\alpha^3$ gives $\alpha = -4^{1/3} \approx -1.587$.
Two-term series. Solving the functional equation with $g \approx 1 + c_4 x^4 + c_8 x^8$ gives $\alpha \approx -1.732$, $c_4 \approx -1.855$, $c_8 \approx 0.278$ (so $g \approx 1 - 1.855x^4 + 0.278x^8$). The accepted value (Briggs 1991, from the literature) is $\alpha \approx -1.690$, so the estimates -1.59, -1.84, -1.73 are within about $6\%, 9\%, 2.5\%$.
Numerical δ. Superstable values for $f = r - x^4$ ($f^{2^n}(0, R_n) = 0$): $R_n = 0, 1, 1.145714, 1.165267, 1.167939, 1.168305, 1$ ($n = 0, \ldots, 7$). Ratios $\frac{R_{n-1} - R_{n-2}}{R_n - R_{n-1}}$: 6.863, 7.453, 7.316, 7.302, 7.289, 7.286, so $\boxed{\delta \approx 7.29}$ for the quartic case (literature: 7.285), noticeably larger than 4.669. ■

10.7.8 *(Intermittency: algebraic version)*

$f(x,r) = -r + x - x^2$.

Solution. **(a)** $f(x) - x = -(r + x^2)$. At $r = 0$: $f(0) = 0$, $f'(0) = 1$, $f''(0) = -2 \neq 0$, $\partial f/\partial r = -1 \neq 0$: a tangent bifurcation. For $r < 0$ there are fixed points $x = \pm\sqrt{-r}$; for $r > 0$ none.
(b) For $r > 0$ small, $f(x) < x$ for all x: the graph lies just below the diagonal near 0, and the cobweb is a long staircase with tiny steps $r + x^2$ through the channel between the curve and the diagonal. So a typical orbit takes many iterations to get through the bottleneck. (Continuum approximation: $\mathrm{d}x/\mathrm{d}n = -(r + x^2)$, so crossing from $x = +a$ to $-a$ takes $\int dx/(r + x^2) \approx \pi/\sqrt{r}$ iterations.)
(c) Near the origin f^2 looks like a rescaled version of f (see (d)); an orbit of f^2 passes through the same bottleneck taking steps of two iterations of f each, so it needs about $\frac{1}{2}N(r)$ iterations of f^2.
(d) $f^2(x) = -r + f - f^2$ with $f = -r + x - x^2$:

$$f^2(x) = -(2r + r^2) + (1 + 2r)x - (2 + 2r)x^2 + O(x^3) \approx -2r + x - 2x^2$$

to leading order in small r. Let $X = 2x$ (so $X' = 2f^2(X/2)$): $F(X) = -4r + X - X^2 = -R + X - X^2$ with $R = 4r$. Thus the second iterate of $f(\cdot, r)$ is, after rescaling, the original map with parameter $4r$. The number of f^2-iterations needed with parameter r equals $N(4r)$, so $\boxed{\frac{1}{2}N(r) \approx N(4r)}$.
(e) Try $N(r) = ar^b$: $\frac{1}{2}ar^b = a(4r)^b = a4^b r^b \Rightarrow 4^b = \frac{1}{2} \Rightarrow b = -\frac{1}{2}$. So $N(r) \propto r^{-1/2}$, in agreement with the direct estimate $\pi/\sqrt{r}$. ■

10.7.9 *(Intermittency: functional version)*

Solution. Exact renormalization of the tangent bifurcation (bring f^2 near the origin back to the form of f by rescaling $x \to \alpha x$) gives the same functional equation $g(x) = \alpha g^2(x/\alpha)$ but with the boundary conditions of a tangency, $g(0) = 0$, $g'(0) = 1$.
(a) Let $g(x) = \dfrac{x}{1 + ax}$. Then $g(g(y)) = \dfrac{y/(1 + ay)}{1 + ay/(1 + ay)} = \dfrac{y}{1 + 2ay}$. With $\alpha = 2$: $\alpha g^2(x/\alpha) = 2 \cdot \dfrac{x/2}{1 + 2a(x/2)} = \dfrac{x}{1 + ax} = g(x)$ ✓, and $g(0) = 0$, $g'(0) = 1$ ✓ for any a.
(b) Near the bottleneck $g(x) \approx x - ax^2$, so a cobweb of g is a staircase with step length $\approx ax^2$ per iteration. The second iterate g^2 takes two such steps at once, so its staircase has steps twice as long. To make g^2 look like g we must stretch x by a factor 2: hence $\alpha = 2$. (This is the same factor $X = 2x$ that appeared in 10.7.8.) ■

10.7.10 *(Algebra for period doubling)*

$f(x) = -(1 + \mu)x + x^2$.

Solution. Let p, q be the 2-cycle ($p + q = \mu$, $pq = -\mu$, from the text) and $s = 1 + \mu$. Expand $p + \eta_{n+1} = f^2(p + \eta_n)$ in powers of η_n:

$$\eta_{n+1} = (f^2)'(p)\,\eta_n + \tfrac{1}{2}(f^2)''(p)\,\eta_n^2 + O(\eta_n^3).$$

Linear coefficient. $(f^2)'(p) = f'(p)f'(q)$ with $f'(x) = 2x - s$:

$$(2p - s)(2q - s) = 4pq - 2s(p + q) + s^2 = -4\mu - 2(1 + \mu)\mu + (1 + \mu)^2 = 1 - 4\mu - \mu^2.$$

Quadratic coefficient. $(f^2)''(x) = f''(f(x))f'(x)^2 + f'(f(x))f''(x)$ with $f'' = 2$, so $\frac{1}{2}(f^2)''(p) = f'(p)^2 + f'(q) = (2p - s)^2 + 2q - s$. Since $q = -sp + p^2$ and $p = -sq + q^2$ we have $p^2 = q + sp$; so

$(2p-s)^2 = 4p^2 - 4sp + s^2 = 4q + s^2$ and

$$\tfrac{1}{2}(f^2)''(p) = 4q + s^2 + 2q - s = 6q + s^2 - s = 6q + \mu + \mu^2.$$

With $q = \frac{\mu - \sqrt{\mu^2+4\mu}}{2}$: $C = 3\mu - 3\sqrt{\mu^2+4\mu} + \mu + \mu^2 = 4\mu + \mu^2 - 3\sqrt{\mu^2+4\mu}$. Hence

$$\eta_{n+1} = (1 - 4\mu - \mu^2)\eta_n + C\eta_n^2 + \cdots, \qquad C = 4\mu + \mu^2 - 3\sqrt{\mu^2+4\mu},$$

confirming (10.7.4) and (10.7.5). ■

10.7.11 *(Cobweb analysis of the recursion for μ_k)*

Solution. The forward iteration is $\mu_k = h(\mu_{k-1})$, $h(\mu) = -2 + \sqrt{6+\mu}$ (obtained by solving $\mu_{k-1} = \mu_k^2 + 4\mu_k - 2$ for the positive root μ_k; the sign printed in (10.7.10) of the text appears to be a typesetting slip, since $\mu_2 = -2 + \sqrt{6} = 0.449$ is required). h is increasing and concave on $\mu \geq -6$. Fixed points: $\mu + 2 = \sqrt{6+\mu} \Rightarrow \mu^2 + 3\mu - 2 = 0$, so $\mu^* = \dfrac{-3+\sqrt{17}}{2} \approx 0.5616 > 0$ (the other root is negative and extraneous). Stability: $h'(\mu^*) = \dfrac{1}{2\sqrt{6+\mu^*}} = \dfrac{1}{2(\mu^*+2)} = 0.195 < 1$, so μ^* is stable (and $0.195 = 1/5.12 = 1/(2\mu^* + 4)$). Cobweb from $\mu_1 = 0$: $h(0) = 0.449 > 0$, so the graph lies above the diagonal for $\mu < \mu^*$ (since $h(\mu) > \mu \iff \mu < \mu^*$ on $\mu \geq 0$); the cobweb is a monotone staircase rising to the right and $\mu_k \uparrow \mu^*$, with $\mu_k = 0,\ 0.4495,\ 0.5396,\ 0.5573,\ 0.5607,\ 0.5614, \ldots$ This μ^* is the onset of chaos, $r_\infty \approx 3 + \mu^* = 3.56$ for the logistic map. ■

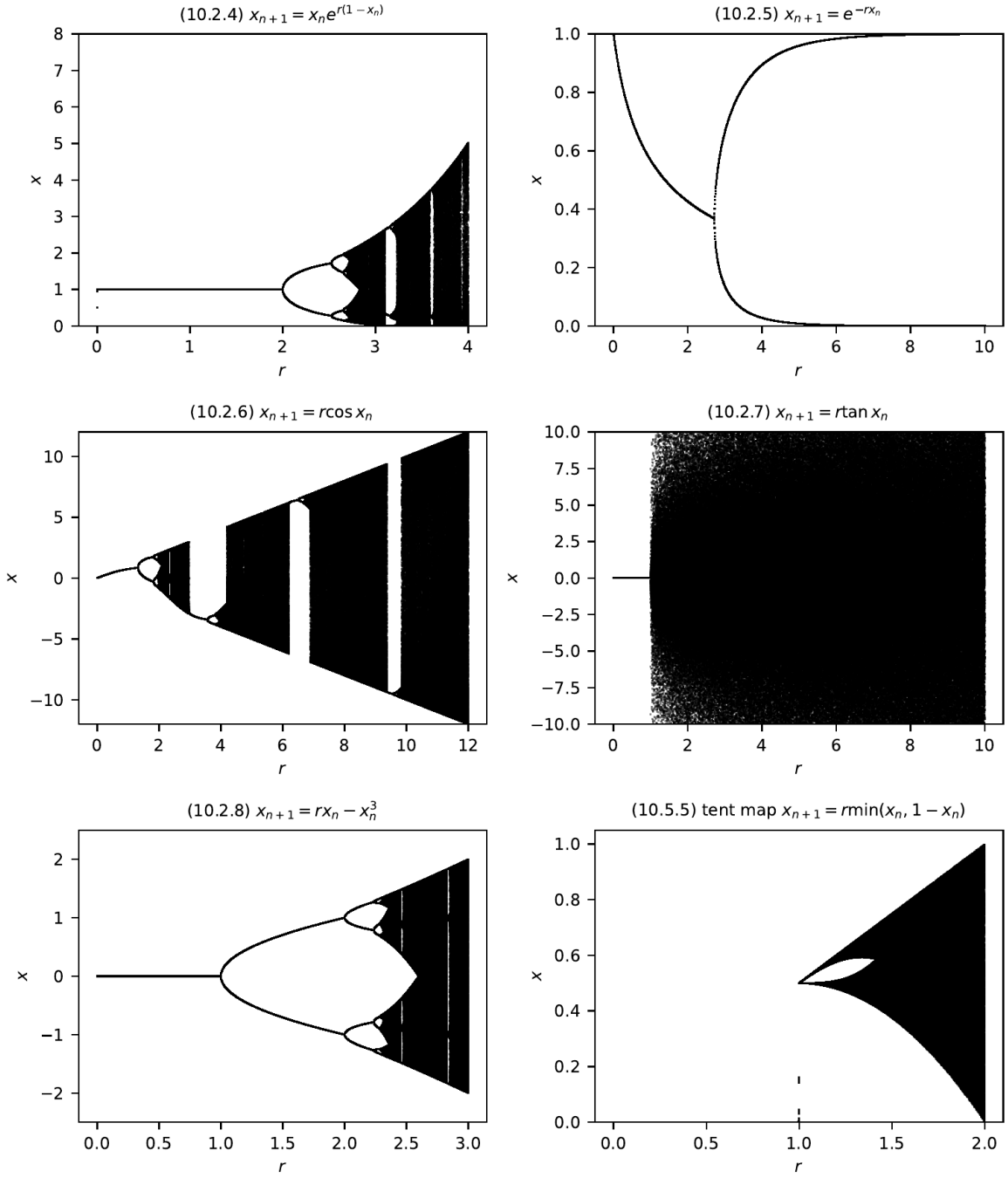


Figure 2: Orbit diagrams. First five panels: Exercises 10.2.4–10.2.8 (the cubic one is plotted from both $x_0 = 0.3$ and $x_0 = -0.3$); last panel: tent map (10.5.5).

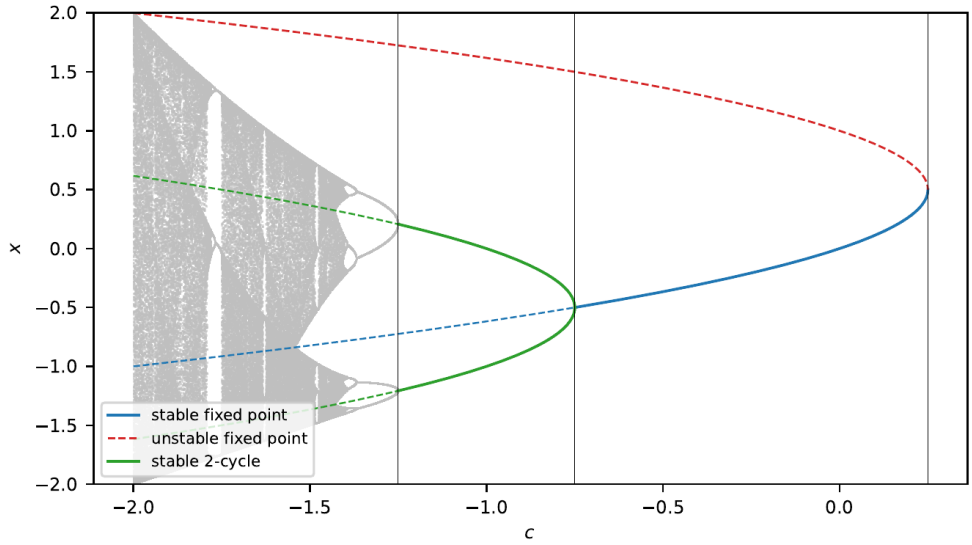


Figure 3: Quadratic map: fixed points, 2-cycle and their stability. Vertical lines at $c = \frac{1}{4}, -\frac{3}{4}, -\frac{5}{4}$.

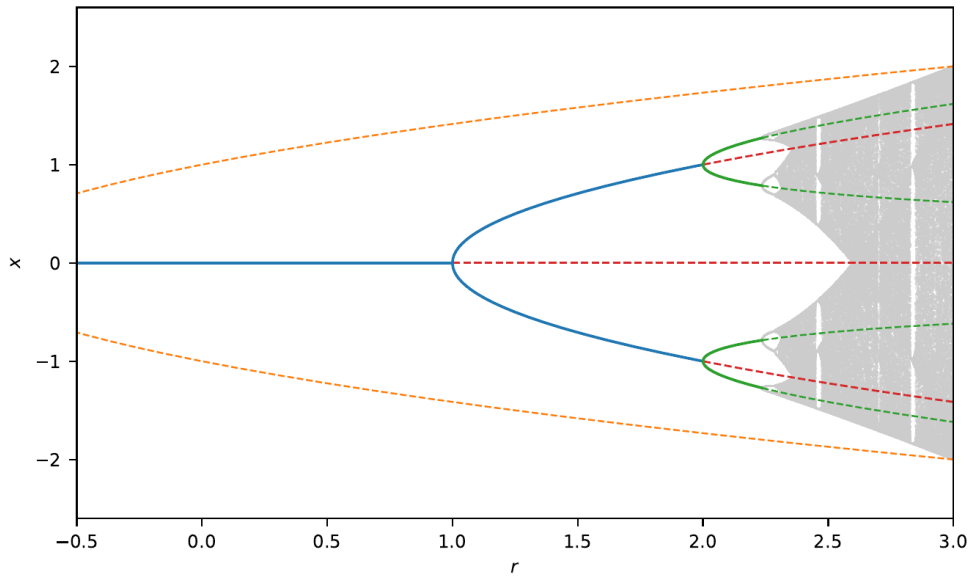


Figure 4: Cubic map. Solid: stable; dashed: unstable. Blue/red: fixed points; green: the 2-cycles $\{p, 1/p\}$; orange: the always-unstable cycle $\{\pm\sqrt{r+1}\}$. Grey: numerical orbits.

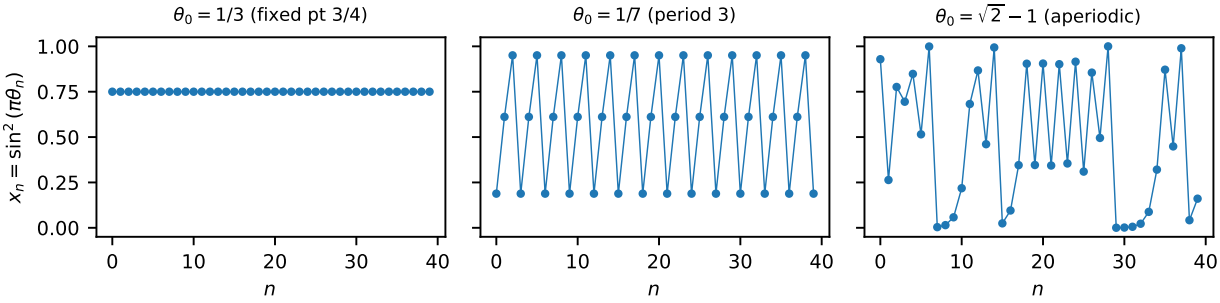


Figure 5: $x_n = \sin^2(\pi\theta_n)$ for three choices of θ_0.

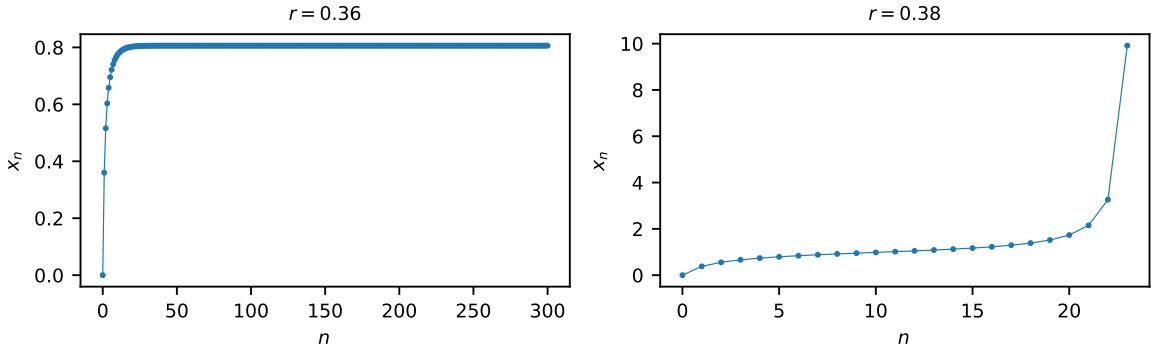


Figure 6: x_n for $r = 0.36$ (left) and $r = 0.38$ (right), $x_0 = 0$ (stopped when $x_n > 5$).

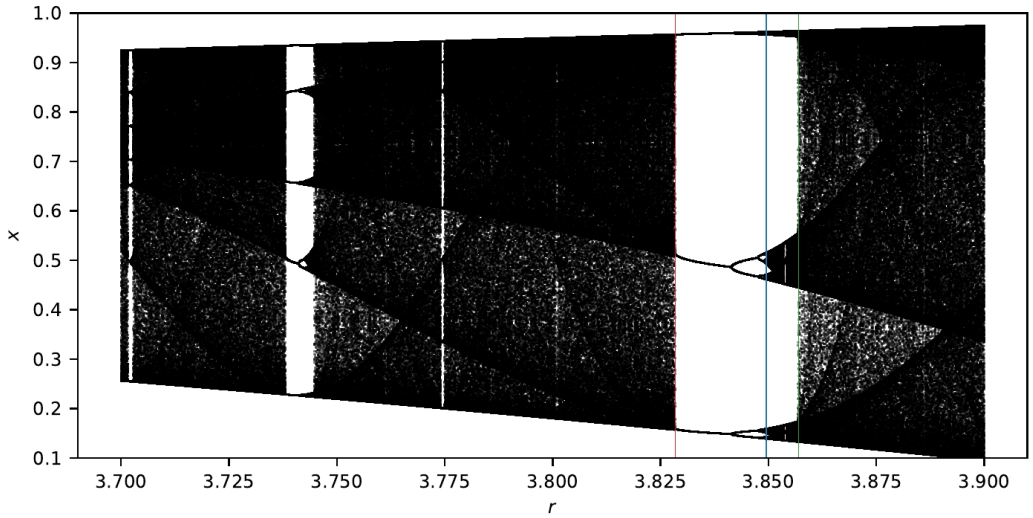


Figure 7: Zoom of the logistic orbit diagram around the period-3 window.

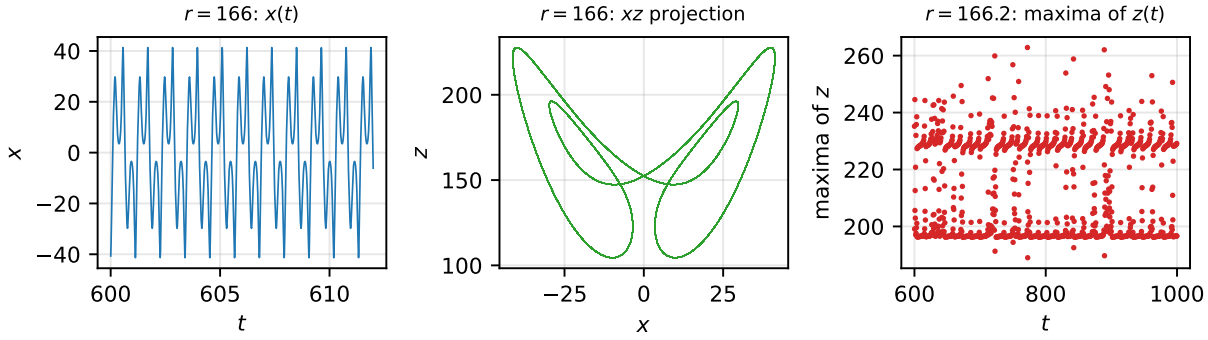


Figure 8: Left, middle: the limit cycle at $r = 166$. Right: successive maxima of z at $r = 166.2$ – most sit on the two laminar levels ≈ 196 and ≈ 228, with bursts of scatter in between.

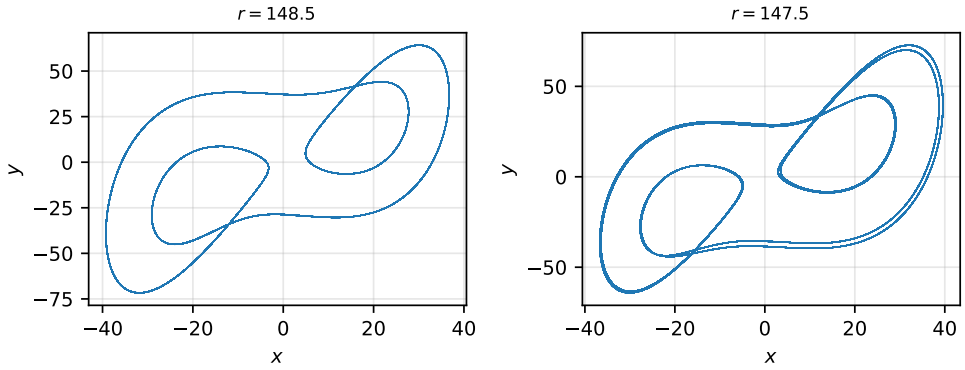


Figure 9: xy projections, $r = 148.5$ (left) and $r = 147.5$ (right).

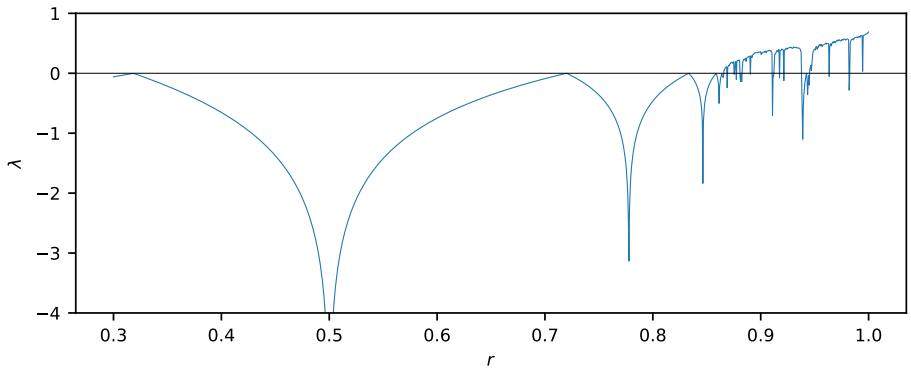


Figure 10: Liapunov exponent of the sine map.

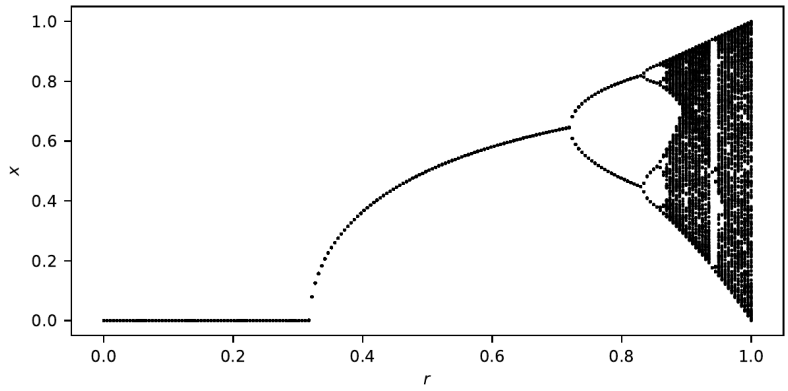


Figure 11: Sine map orbit diagram (200 values of r).

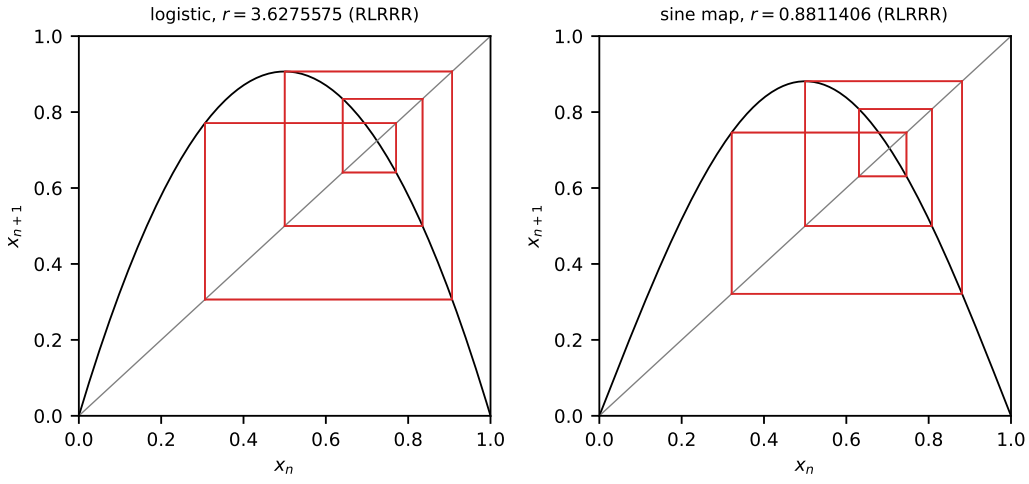


Figure 12: Superstable 6-cycles: logistic ($r = 3.6275575$) and sine map ($r = 0.8811406$).

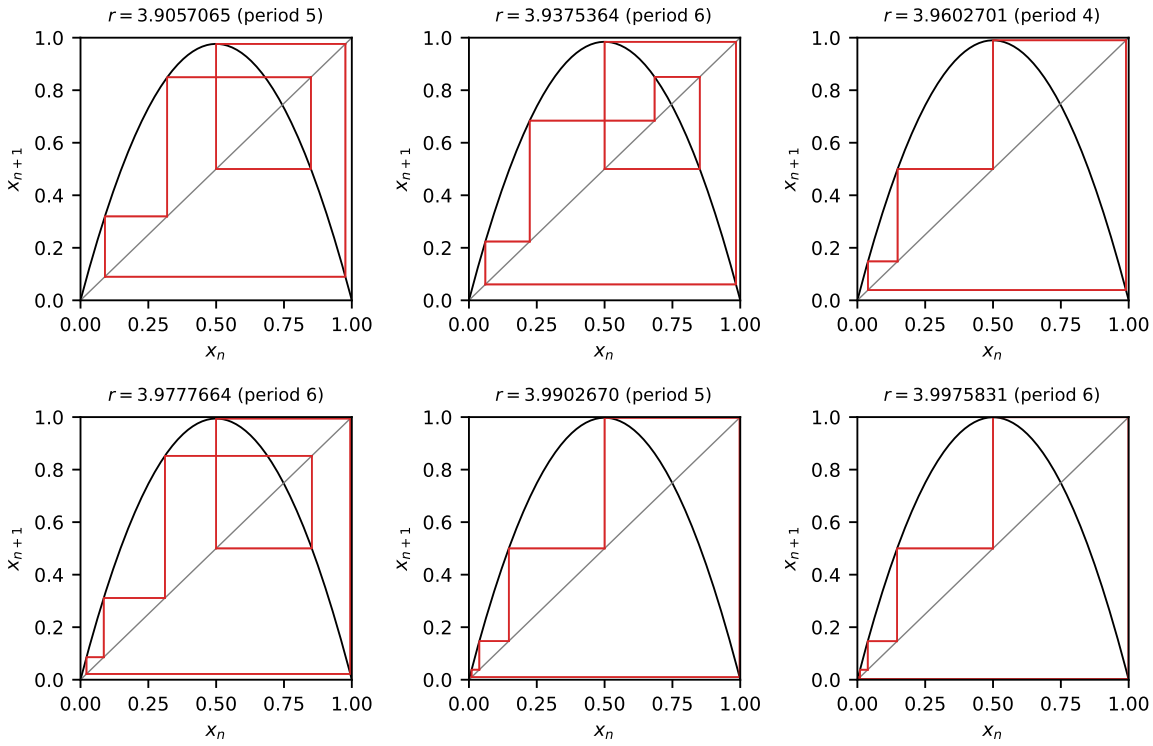


Figure 13: Cobwebs for the superstable cycles of periods $5, 6, 4, 6, 5, 6$ (logistic map).